

FACE ATTRIBUTE RECOGNITION USING DEEP LEARNING MODELS

Comparative Analysis of MLP, CNN, and Vision Transformer
for Eyeglasses Detection and Facial Similarity Matching

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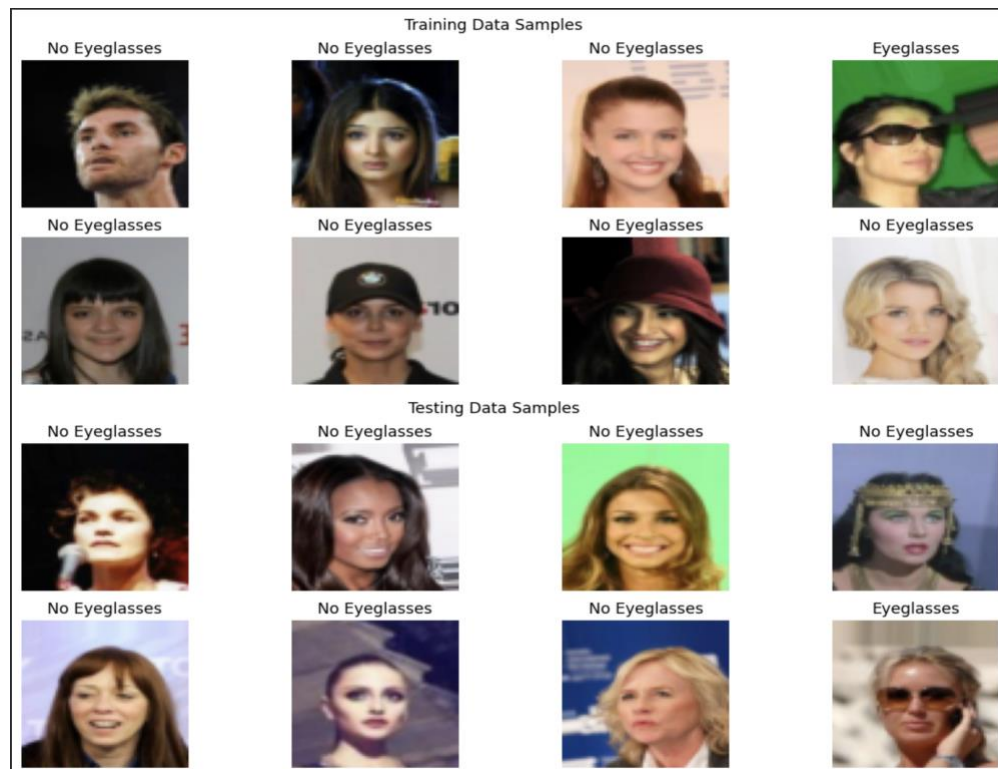
PROJECT SUMMARY

Motivation

Facial attribute recognition is an important Computer Vision task used in applications such as facial verification, assistive technology, smart surveillance, content filtering, and social media tagging. Modern AI systems must be able to automatically recognize facial features and attributes from images under different lighting conditions, poses, and image qualities.

This project focuses on detecting the “Eyeglasses” attribute using Deep Learning techniques on the CelebA dataset. The problem is challenging because the dataset is highly imbalanced:

- 93.4% images without glasses
- 6.6% images with glasses



This imbalance makes traditional accuracy metrics misleading and creates difficulty in correctly detecting minority class samples.

Main Idea

The main goal of the project was to compare different Deep Learning architectures for facial attribute recognition and evaluate which model performs best for eyeglasses detection.

Three models were implemented and compared:

- Multi-Layer Perceptron (MLP)

- Convolutional Neural Network (CNN)
- Vision Transformer (ViT-style)

The project also explored several improvement techniques discussed during the course:

- Weighted CrossEntropyLoss for class imbalance handling
- Threshold tuning for improving Recall and F1 Score
- Gaussian noise robustness testing

The CNN architecture became the primary focus because convolution layers preserve spatial information and learn important local image features such as edges and glasses frame structures.

Results

The CNN achieved the best overall performance among all tested models.

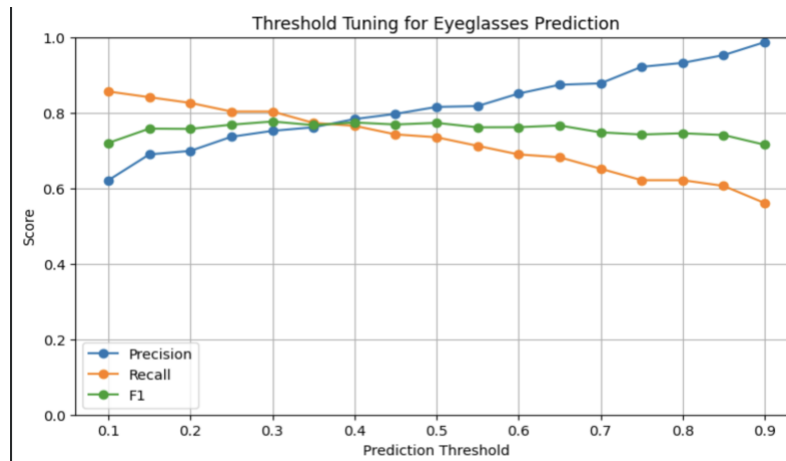
Final Model Comparison:				
	Accuracy	Precision	Recall	F1
CNN Image Recognition	0.9715	0.815126	0.734848	0.772908
MLP Neural Network	0.9375	0.518919	0.727273	0.605678
Transformer	0.8770	0.332353	0.856061	0.478814
Best model: CNN Image Recognition				

Final CNN Performance:

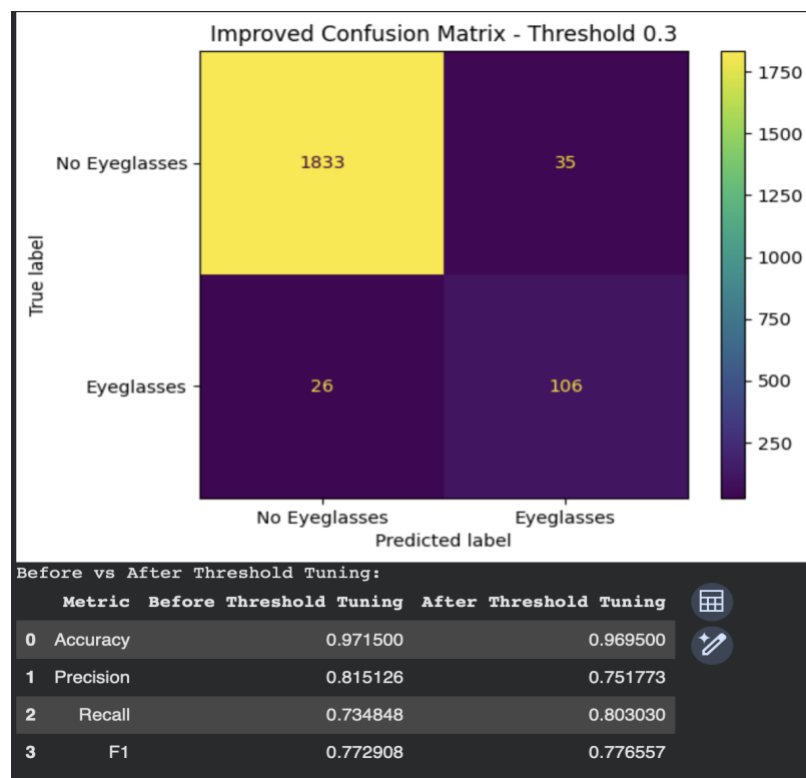
- Accuracy = 97.15%
- Precision = 81.5%
- Recall = 73.5%
- F1 Score = 77.3%



Threshold tuning further improved minority class detection by lowering the prediction threshold from 0.50 to 0.30:



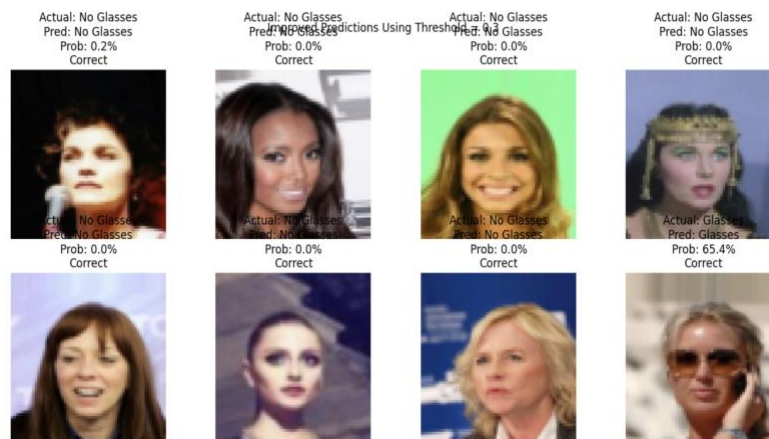
- Recall improved from 73.5% $\rightarrow$ 80.3%
- F1 Score improved from 77.3% $\rightarrow$ 77.7%



The experiments demonstrated several important findings:

- CNNs outperform MLPs for image classification because they preserve spatial information
- Vision Transformers require significantly larger datasets for effective training
- Weighted loss functions are essential for imbalanced datasets

- Explainable AI methods confirmed that the CNN focused on meaningful facial regions such as the eyes and glasses frames
- Deep learning models are highly sensitive to image noise and corruption



The project also includes a facial attribute similarity system that searches for visually similar faces based on facial regions such as eyes, nose, mouth, and hair using RGB histogram features and cosine similarity matching.

Best attribute match found in first 500 images
 Best overall match: 82.75%
 This is not person identification. It is only attribute similarity.

	Feature	Image A	Image B	Match %
0	Hair visual similarity	Region feature	Region feature	58.49
1	Eyes visual similarity	Region feature	Region feature	88.25
2	Nose visual similarity	Region feature	Region feature	95.41
3	Mouth visual similarity	Region feature	Region feature	67.29
4	Full_face visual similarity	Region feature	Region feature	87.75
5	Hair color	Brown Hair	Other/Unknown	0.00
6	Hair type	Other/Unknown	Other/Unknown	100.00
7	Nose feature	Other/Normal Nose	Other/Normal Nose	100.00
8	Eye feature	Other/Normal Eyes	Other/Normal Eyes	100.00
9	Age group	Young	Young	100.00
10	Eyeglasses	No Eyeglasses	No Eyeglasses	100.00
11	Smile	Not Smiling	Not Smiling	100.00
12	Beard	No Beard	No Beard	100.00
13	Hat	No Hat	No Hat	100.00

Best Attribute Match: 82.75%

Query Image	Best Attribute Match

LITERATURE REVIEW — RELATED METHODS

This project applies several Artificial Intelligence and Deep Learning concepts covered throughout the course, including Neural Networks, Computer Vision, Transformers, Explainable AI (XAI), Optimization, and Robustness Testing. The project combines these concepts to build and improve a facial attribute recognition system using the CelebA dataset.

1. Neural Networks and Multi-Layer Perceptron (MLP)

One of the first concepts introduced in the course was the Artificial Neural Network (ANN) and Multi-Layer Perceptron (MLP). MLPs consist of fully connected layers where neurons learn patterns by adjusting weights through backpropagation and gradient descent.

In this project, an MLP model was implemented as the baseline architecture. The input face image of size $64 \times 64 \times 3$ was flattened into a vector of 12,288 features and passed through multiple dense layers with ReLU activation and dropout regularization. This helped demonstrate how traditional neural networks perform on image classification tasks. However, because MLPs treat images as flat vectors, the model struggled to learn spatial patterns such as eyeglass frame edges and facial structures.

2. Convolutional Neural Networks (CNNs)

The course introduced Convolutional Neural Networks as one of the most important methods in Computer Vision. CNNs use convolution filters to automatically extract local visual features such as edges, curves, textures, and shapes while preserving spatial information.

In this project, CNN was the main architecture used for facial attribute recognition. The model contained four convolutional blocks with Batch Normalization and MaxPooling layers. The CNN learned meaningful visual features from the eye and glasses regions of the face and achieved the highest overall performance among all tested models.

The CNN achieved:

- Accuracy = 97.15%
- Precision = 81.5%
- Recall = 73.5%
- F1 Score = 77.3%

The strong performance demonstrated how convolution operations improve image classification by learning local spatial patterns directly from raw images.

3. Vision Transformer (ViT)

Another major topic discussed in the course was Transformers and self-attention mechanisms. Vision Transformers divide images into smaller patches and use self-attention to learn global relationships between image regions.

In this project, a lightweight ViT-style architecture was implemented using:

- 8×8 image patches
- 128-dimensional embeddings
- 2 Transformer encoder layers
- 4 attention heads

The Transformer model was able to learn global image relationships, but it underperformed compared to CNN because the dataset size was relatively small (8,000 images). This experiment demonstrated an important concept discussed in class: Transformers generally require much larger datasets and computational resources than CNNs.

4.Optimization and Hyperparameter Tuning

The course also discussed optimization methods and hyperparameter tuning in Deep Learning systems. Hyperparameters such as learning rate, batch size, epochs, dropout, and optimizers strongly influence model performance.

In this project:

- Adam optimizer was used for faster convergence
- Learning rate = 0.001
- Batch size = 64
- Epochs = 10
- Dropout layers were added to reduce overfitting

The project also explored threshold tuning. Instead of using the default probability threshold of 0.50, thresholds between 0.10 and 0.90 were tested. The best threshold was found to be 0.30, which improved Recall from 73.5% to 80.3% and slightly improved the F1 Score.

This demonstrated how adjusting decision boundaries can improve minority class detection in imbalanced datasets.

5.Class Imbalance Handling

Class imbalance is a common issue in machine learning where one class appears much more frequently than another. The course discussed how imbalanced datasets can produce misleadingly high accuracy while failing to detect minority classes.

In this project:

- 93.4% of images contained no eyeglasses
- Only 6.6% contained eyeglasses

To address this issue, Weighted CrossEntropyLoss was used. Higher loss weights were assigned to the minority “Glasses” class so that mistakes on those samples were penalized more heavily.

This significantly improved Recall and prevented the model from predicting only the majority class.

6. Robustness Testing and Noise Analysis

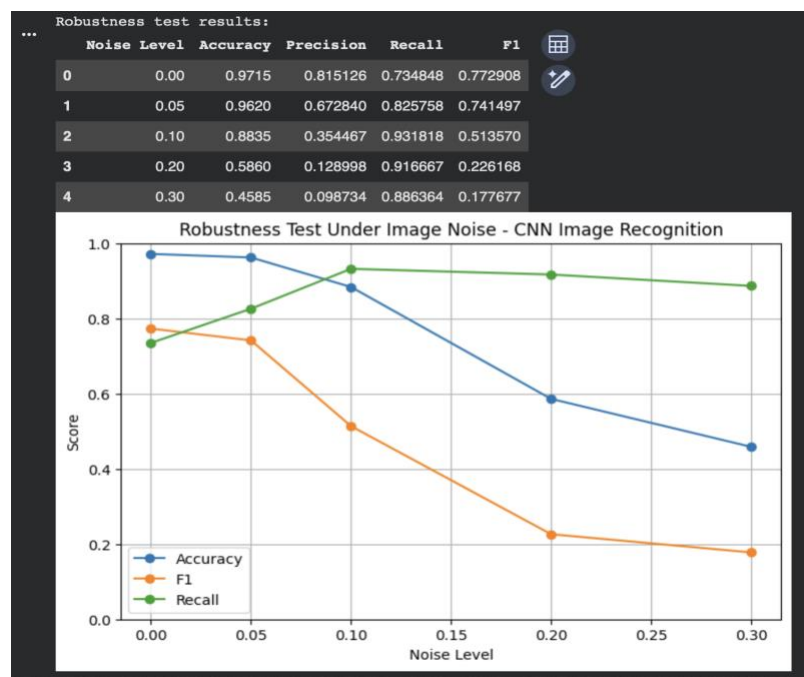
The course also introduced robustness and reliability testing in AI systems. Real-world images often contain blur, distortion, or noise, which can reduce model accuracy.

In this project, Gaussian noise was added to test images at different levels:

- 0.00
- 0.05
- 0.10
- 0.20
- 0.30

The experiments showed that CNN performance degraded significantly as noise increased. At heavy noise levels, the F1 Score dropped from 77.3% to 17.8%.

This experiment demonstrated the importance of robustness testing in real-world AI systems and highlighted the sensitivity of deep learning models to corrupted image inputs.



7. Face Similarity and Content-Based Image Retrieval

Another concept related to this project is facial similarity matching and Content-Based Image Retrieval (CBIR). Instead of identifying a person's identity, similarity systems compare visual facial features such as eyes, nose, hair, and mouth regions to retrieve visually similar faces.

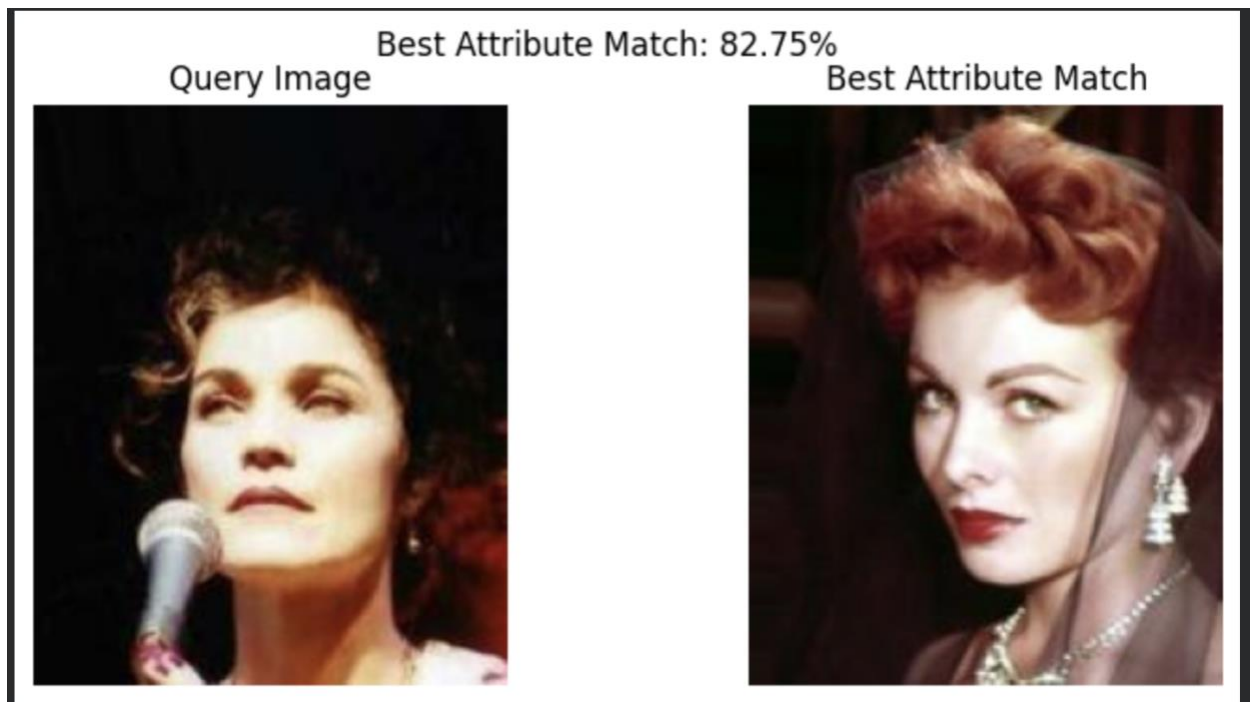
In this project, a facial similarity module was implemented after the attribute classification stage. The system extracted RGB histogram features from multiple facial regions and compared them

using cosine similarity. This allowed the model to find visually similar faces from a dataset of candidate images.

The similarity system produced:

- Overall facial similarity percentage
- Region-wise similarity scores
- Eyes similarity
- Nose similarity
- Hair similarity
- Mouth similarity

The best matching faces achieved similarity scores above 80%, demonstrating how feature extraction and similarity metrics can be combined with Deep Learning systems for facial comparison tasks.



APPROACH USED

1.Dataset Collection

The project used the CelebA (CelebFaces Attributes Dataset) dataset from HuggingFace.

Dataset Source:

<https://huggingface.co/datasets/nateraw/celeba>

The CelebA dataset contains:

- 200,000+ celebrity face images
- 40 binary facial attributes
- Large variations in lighting, pose, facial expression, and background

For this project:

- 8,000 images were selected
- Target attribute: Eyeglasses
- Binary classification:
 - 1 = Glasses
 - 0 = No Glasses

The dataset was randomly shuffled using seed = 42 to ensure reproducibility.

2.Data Preprocessing

Several preprocessing steps were applied before training the models.

Image Resize

All images were resized to:

64 × 64 pixels

Reason:

- Reduces computational cost
- Standardizes image dimensions
- Speeds up training

Normalization

Pixel values were normalized between -1 and +1.

Benefits:

- Faster convergence
- Stable gradient updates
- Improved optimization performance

Data Augmentation

RandomHorizontalFlip was applied during training.

Purpose:

- Reduce overfitting
- Improve generalization
- Simulate additional training samples

Dataset Split

- Training set = 6,000 images
- Test set = 2,000 images

3.Handling Class Imbalance

The dataset was highly imbalanced:

- 93.4% No Glasses
- 6.6% Glasses

Without correction, the models predicted mostly the majority class.

To solve this issue, Weighted CrossEntropyLoss was used.

Class weights:

- No Glasses = 0.535
- Glasses = 5.29

This increased the penalty for misclassifying minority class samples and significantly improved Recall performance.

PyTorch Loss Function:

<https://pytorch.org/docs/stable/generated/torch.nn.CrossEntropyLoss.html>

4.Deep Learning Models Used

A. Multi-Layer Perceptron (MLP)

The MLP model was implemented as a baseline neural network.

Architecture:

- Input size = 12,288
- Linear(12288 $\rightarrow$ 512)
- ReLU activation
- Dropout(0.30)
- Linear(512 $\rightarrow$ 128)

- ReLU activation
- Dropout(0.20)
- Linear(128 $\rightarrow$ 2)

The MLP flattened images into vectors and processed them through fully connected layers.

Limitation:

MLPs do not preserve spatial image information, reducing their ability to learn facial structures.

B. Convolutional Neural Network (CNN)

CNN was the best-performing model in the project.

Architecture:

- Conv2D(3 $\rightarrow$ 32) + BatchNorm + MaxPooling
- Conv2D(32 $\rightarrow$ 64) + BatchNorm + MaxPooling
- Conv2D(64 $\rightarrow$ 128) + BatchNorm + MaxPooling
- Conv2D(128 $\rightarrow$ 256) + AdaptiveAveragePooling
- Fully Connected 256 $\rightarrow$ 128
- Dropout(0.30)
- Output layer 128 $\rightarrow$ 2

CNNs preserve spatial information and learn important visual features such as:

- edges
- textures
- glasses frames
- facial boundaries

CNN achieved the highest overall performance.

PyTorch CNN Documentation:

<https://pytorch.org/docs/stable/generated/torch.nn.Conv2d.html>

C. Vision Transformer (ViT-style)

A lightweight Vision Transformer architecture was also implemented.

Architecture:

- 8 $\times$ 8 image patches
- 128-dimensional embeddings
- Positional encoding
- 2 Transformer encoder layers
- 4 attention heads
- Mean pooling
- Linear classifier

The Transformer learned global relationships between image patches using self-attention.

However, performance was weaker due to limited dataset size.

Vision Transformer Paper:

<https://arxiv.org/abs/2010.11929>

4. Model Training

All models were trained using the same training configuration for fair comparison.

Training Hyperparameters:

- Optimizer = Adam
- Learning Rate = 0.001
- Epochs = 10
- Batch Size = 64

Adam Optimizer:

<https://pytorch.org/docs/stable/generated/torch.optim.Adam.html>

The models learned by minimizing CrossEntropyLoss through backpropagation and gradient descent.

5. Threshold Tuning

The CNN output probabilities using Softmax activation.

Default threshold:

0.50

Problem:

The model became too conservative and missed many true eyeglasses samples.

Thresholds from 0.10 to 0.90 were tested in steps of 0.05.

Best threshold:

0.30

Results after tuning:

- Recall improved from 73.5% → 80.3%
- F1 Score improved from 77.3% → 77.7%

This demonstrated how threshold tuning improves minority class detection.

6. Explainable AI (XAI)

Gradient-based saliency maps were used to visualize which image regions influenced predictions most strongly.

The saliency maps highlighted:

- eyes
- glasses frames
- nearby facial structures

This confirmed that the CNN learned meaningful visual features rather than background artifacts.

PyTorch Autograd Documentation:

<https://pytorch.org/docs/stable/autograd.html>

7.Noise Robustness Testing

Gaussian noise was added to test images to evaluate model robustness under corrupted image conditions.

Noise levels tested:

- 0.00
- 0.05
- 0.10
- 0.20
- 0.30

Results showed significant performance degradation as noise increased.

This experiment demonstrated the sensitivity of deep learning systems to image corruption and noisy inputs.

8.Evaluation Metrics

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score

F1 Score was considered the most important metric because the dataset was highly imbalanced.

Final CNN Performance:

- Accuracy = 97.15%
- Precision = 81.5%
- Recall = 73.5%
- F1 Score = 77.3%

9. Technologies and Libraries Used

Programming Language:

- Python

Deep Learning Framework:

- PyTorch

Libraries:

- NumPy
- OpenCV
- Matplotlib
- HuggingFace Datasets
- Torchvision

PyTorch Documentation:

<https://pytorch.org/docs/stable/index.html>

HuggingFace Datasets:

<https://huggingface.co/docs/datasets/index>

Link to the Project code

https://github.com/Jashwanth248/ai-hw-Spring-2026-nn/blob/main/MRP_Similar_Face_.ipynb

YouTube Link

<https://youtu.be/YaTw-ACGS0U>